La Scuola d'Italia / Huson / Physics Nickname:
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2.2 Classwork: Average and Instantaneous Velocity

Reference (provided). average speed $= \dfrac{\text{distance}}{\text{time}}$ average velocity $\bar{v} = \dfrac{\text{displacement}}{\text{time}}$
1 h = 3600 s

1. Swimming world records are set in a 50 m pool. The race distances are exact, so give four significant figures.

   (a) The 50 m freestyle record is 20.91 s, one length of the pool. Find the swimmer's average speed and average velocity.

   (b) The 100 m freestyle record is 46.40 s, two lengths: down the pool and back to the starting wall. Find the swimmer's average speed and average velocity.

2. The hallway walk from 2.1, now with a stopwatch. Take east as positive.

Leg	Displacement	Time	Average velocity
1	8.0 m east	4.0 s	
2	5.0 m west	5.0 s	
3	7.0 m west	5.0 s	

   (a) Fill in the average velocity of each leg. Give each one a sign.

   (b) For the whole walk, find the average speed and the average velocity. Why is one so much smaller than the other?

3. The fastest Frecciarossa trains run from Roma Termini to Milano Centrale, about 570 km by rail, in 2 h 59 min without stopping.

   (a) Find the train's average speed in km/h.

   (b) On the way, the display in the carriage reads 300 km/h. Is that an average or an instantaneous speed? Give one reason the average is lower.

4. **Instantaneous velocity.** A cart is released from rest at the top of a tilted Vernier track. Graphical Analysis gives its position at these times.

t (s)	1.00	1.05	1.10	1.50	2.00
x (m)	0.250	0.276	0.303	0.563	1.000

(a) Find the average velocity over each time interval.

Interval	Δt (s)	Δx (m)	$\bar{v}$ (m/s)
1.00 s to 2.00 s			
1.00 s to 1.50 s			
1.00 s to 1.10 s			
1.00 s to 1.05 s			

(b) As the interval gets shorter, what value is $\bar{v}$ getting close to? That is the cart's instantaneous velocity at $t = 1.00$ s.

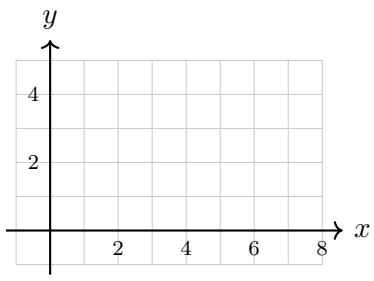
(c) Why can we not keep shrinking the interval to get a better answer from this data?

5. A soccer player runs a drill: 6.0 m east in 2.0 s, then 4.0 m north in 2.0 s, then 6.0 m west in 3.0 s. East is $+x$ and north is $+y$.

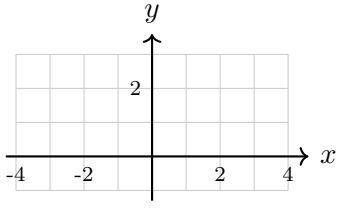
(a) On the path grid, draw the drill as three displacement vectors, each starting where the last one ended. Number them 1, 2, 3.

(b) Find the average velocity of each leg, with its direction. Draw each one as an arrow from the origin of the velocity grid.

(c) For the whole drill, find the average speed and the average velocity. The path grid shows where the player finishes.



path: 1 square = 1 m



velocity: 1 square = 1 m/s